SQuADs Adversaries

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Abstract

Pre-trained models often demonstrate high performance on benchmark datasets, such as SQuAD for question-answering tasks. However, this raises the question: are they truly solving the underlying tasks or merely exploiting dataset artifacts?

We investigate this by introducing adversarial examples after training the ELECTRA-small-discriminator model. Our results reveal significant performance degradation, suggesting that these models primarily memorize answer patterns rather than effectively reasoning at inference time. Detailed analysis highlights subsets of data, particularly numeric and named entities, as primary sources of performance issues on the base SQuAD training set.

Attempts to address these issues through fine-tuning on subsets proved ineffective. As an alternative, we applied the UnifiedQA model for post-processing, achieving substantial performance improvements. These findings underscore the need for robust evaluation methodologies to ensure models go beyond surface-level learning and truly comprehend the tasks they are designed to solve.

1 Introduction

Despite their high performance on benchmark datasets like SQuAD, pre-trained models often fail dramatically when evaluated on adversarial examples. Adversarial questions—crafted to exploit vulnerabilities in model understanding—reduces performance by 66.18 EM / 64.76 F1 in the most extreme case (see Table 1).

Data	EM	F1
Base SQuAD	77.08	85.12
Adversary dBERT	10.90	20.36
Adversary dBiDAF	26.50	38.53
Adversary dRoBERTa	12.70	20.94

Table 1. Adversarial QA Validation

Why This is Important: Question-answering systems play a pivotal role in critical domains such as customer support and healthcare, where timely and accurate responses are essential. Ensuring both efficiency and robustness is crucial, as failures can result in misleading or harmful information, and slow runtimes hinder practical deployment. Achieving high performance without compromising scalability is key to their real-world applicability.

Why This is Challenging: Developing effective question-answering systems requires nuanced reasoning—integrating information across sources, resolving contextual ambiguities, and handling complex dependencies. Using advanced models like UnifiedQA for direct training offers strong reasoning capabilities but is computationally intensive, making it impractical for large-scale or real-time applications. Balancing accuracy and runtime efficiency poses a significant challenge.

Previous Solutions and Their Limitations: Existing approaches, such as training state-of-the-art models like UnifiedQA from scratch or fine-tuning on adversarial datasets, improve performance but are computationally expensive. For example, training UnifiedQA directly on SQuAD alone would require over 13 days (approximately with a NVIDIA RTX4050 GPU) limiting its scalability. These methods also fail to provide practical solutions for efficient application deployment, as they are focused primarily on prediction accuracy without considering system runtime efficiency.

Our Approach: Instead of using UnifiedQA as the base trained model, we leveraged the more efficient ELECTRA model for answer generation and then applied UnifiedQA in a post-processing step to refine its answers. This approach drastically reduced the overall runtime to under 1 hour (including training and post-processing), compared to the 13 days required for training UnifiedQA directly, while maintaining similar performance. By combining the computational efficiency of ELECTRA with the reasoning strengths of UnifiedQA, this hybrid methodology achieved a balance of scalability and accuracy. This approach demonstrates that robust post-processing can match the performance of fully trained models.

Key Results and Limitations: Our method achieved significant improvements in runtime efficiency while maintaining performance comparable to directly training UnifiedQA. This hybrid approach underscores the importance of leveraging complementary strengths of models to optimize for both accuracy and efficiency. However, it relies on high-quality adversarial datasets for evaluation and does not fully address fundamental architectural limitations. Future work should focus on refining hybrid pipelines and exploring architectural innovations to enhance reasoning, generalization, and scalability further.

2 Implementation

Initially we attempted Dataset Cartography (Swayamdipta et al., 2020) with ambiguous and hard subsets of SQuAD being fine tuned to improve performance on both baseline and adversarial validation sets. This attempt proved ineffective, but is detailed in sections 3.1-3.2.

After analyzing common answer mistakes, we enhanced the quality of answers generated by the base ELECTRA model through the introduction of UnifiedQA (Khashabi et al., 2020) as a post-processing step. This hybrid approach combined the computational efficiency of ELECTRA with the reasoning strength of UnifiedQA. This substantially improved performance therefore we document it's implementation process as follows, and provide accuracy and performance improvement details in section 3.3.

2.1 Training Parameters

The training process for the base model (ELECTRA-small-discriminator) was conducted using the following key parameters:

```
SEED=42
MODEL_NAME="electra-small-discriminator"
DATASET_NAME="squad"
PER_DEVICE_TRAIN_BATCH_SIZE=12
LEARNING_RATE=3e-5
NUM_EPOCHS=3
MAX_SEQ_LENGTH=384
DOC_STRIDE=128
```

These parameters ensured reproducibility and efficiency during training.

2.2 Pipeline Design

Predictions from the ELECTRA model served as input to UnifiedQA. For each question, context, and initial answer, UnifiedQA refined the answer to improve reasoning accuracy, particularly for challenging cases.

2.3 Batch Processing Optimization

Inputs were processed in batches of size 8 to maximize GPU utilization. UnifiedQA was deployed in evaluation mode, avoiding additional gradient computations and further improving runtime efficiency.

2.4 Input Construction

Inputs for UnifiedQA were structured as:

```
question: [question]
context: [context]
answer: [ELECTRA answer]
```

This ensured consistency and completeness during refinement.

2.5 Decoding and Mapping

UnifiedQA outputs were generated using beam search (beam size = 4), ensuring high-quality results. Refined answers were then mapped back to their respective examples for evaluation.

2.6 Validation

We validated SQuAD for task based performance on adversarial datasets (Bartolo et al., 2020) to assess generalizability and robustness of our results and only training our models on the base SQuAD dataset.

3 Analysis

3.1 Dataset Cartography

We map and diagnose datasets with training dynamics using dataset cartography, with plots of confidence versus variability across epochs shown below in Figure 1.

Our preliminary approach was to attempt fine-tuning on ambiguous and hard-to-learn subsets. Fine-tuning on hard and ambiguous subsets does not yield signficant performance measure improvements, and across all perfmuations we find performance degradation in nearly all cases (see Appendix Table 2 for details).

Subsequent analysis of common answer errors shows subsets of named entities and numeric categories being common incorrect answers on the base squad dataset (Figure 2). We then employ fine-tuning on Named Entity and Numeric subsets of the data. Again, across all perfmuations we find performance degradation in nearly all cases (see Appendix Table 3 for details).

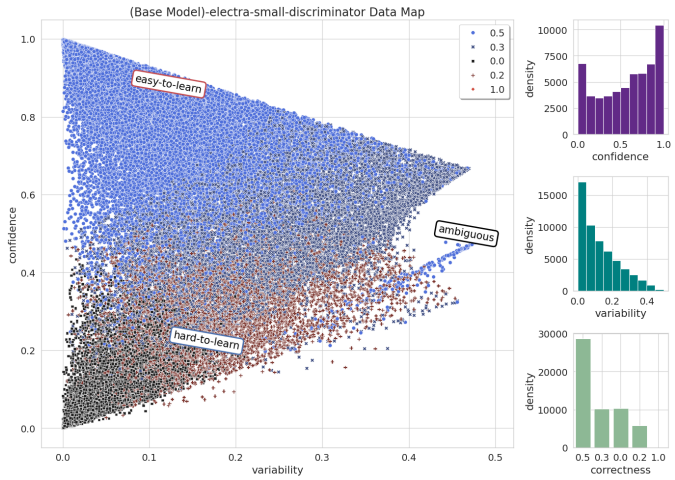

Figure 1. ELETRA-small-discriminator Dataset Cartography on SQuAD datasets.

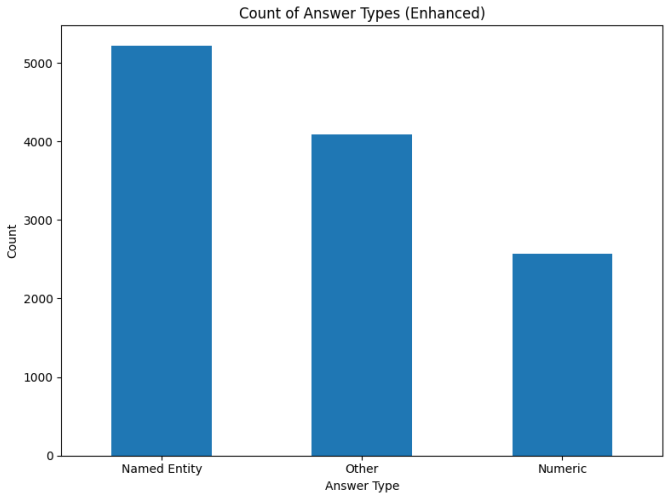

Figure 2. Named Entity and Numeric answer types are frequently predicted incorrectly. Other types are not systematic and contain fine-grained errors that were not easily categorized.

3.2 Hyperparameter Optimization

Optimizing the learning rate revealed that different permutations of fine-tuning steps converge to similar performance outcomes. Specifically, Base + Amb + Hard and Base + Hard + Amb, as well as Base + Num + Nam and Base + Nam + Num, achieved equivalent results. This suggests that effective hyperparameter tuning, particularly for the learning rate, can mitigate order-dependent effects in the fine-tuning process (see Appendix for details).

3.3 Post-Processing Answers

Data	EM	F1	Δ EM / FM
Base SQuAD	83.19	90.48	6.11 / 5.36
Adversary dBERT	23.20	34.94	12.30 / 14.58
Adversary dBiDAF	44.00	56.99	17.50 / 18.46
Adversary dRoBERTa	18.30	28.25	5.60 / 6.26

Table 2. Post-Processing QA Validation

Post-processing the base model's predictions with UnifiedQA led to substantial performance improvements across multiple metrics. As shown in the table, Base + UnifiedQA improved Exact Match (EM) by 6.11 points and F1 by 5.36 points compared to the base model alone. Additionally, significant gains were observed in harder subsets: numeric questions saw a 17.50-point improvement in EM and 18.46 points in F1, while ambiguous and named entity subsets gained 12.30 and 14.58 points, respectively. This highlights the effectiveness of UnifiedQA in refining answers, especially for challenging subsets, without requiring extensive re-training or model architecture changes.

sectionConclusion

Our study demonstrates that leveraging UnifiedQA for post-processing can significantly enhance the performance of a base model like ELECTRA on adversarial and challenging subsets. This hybrid approach achieves competitive performance while reducing runtime from 13 days to under 1 hour. The proposed method effectively combines computational efficiency with enhanced reasoning capabilities, providing a viable solution for real-world question-answering systems.

However, reliance on high-quality adversarial datasets and architectural limitations highlight areas for future improvement. This work emphasizes the potential of hybrid approaches in overcoming limitations of individual models.

4 Future Work

- Explore alternative reasoning models for post-processing to improve generalization and scalability.
- Develop hybrid pipelines that integrate reasoning during training, rather than post-processing alone.
- Resolve additional issues within the Other Answer Type category.

5 Appendix A

Data	Base	EM	F1
Base SQuAD	+Amb.	77.14	84.84
Adversary dBERT	+Amb.	10.50	19.89
Adversary dBiDAF	+Amb.	27.00	38.18
Adversary dRoBERTa	+Amb.	13.20	21.03
Base SQuAD	+Hard	74.83	83.73
Adversary dBERT	+Hard	11.10	20.04
Adversary dBiDAF	+Hard	25.70	37.47
Adversary dRoBERTa	+Hard	11.60	21.08
Base SQuAD	+Amb +Hard	75.24	83.89
Adversary dBERT	+Amb +Hard	12.10	20.83
Adversary dBiDAF	+Amb +Hard	27.00	37.70
Adversary dRoBERTa	+Amb +Hard	12.00	21.02
Base SQuAD	+Amb +Hard	77.13	84.93
Adversary dBERT	+Amb +Hard	11.90	20.16
Adversary dBiDAF	+Amb +Hard	27.30	38.42
Adversary dRoBERTa	+Amb +Hard	12.70	20.94

Table 3. Ambiguous + Hard Subsets Fine-tuning the base model with dataset cartography ambigious (Amb.) and hard (Hard) subsets does not yield consistent performance level increases over base model.

Data	Base	EM	F1
Base SQuAD	+NE	75.18	84.03
Adversary dBERT	+NE	10.30	20.13
Adversary dBiDAF	+NE	25.80	36.99
Adversary dRoBERTa	+NE	12.40	21.63
Base SQuAD	+Num	72.93	82.26
Adversary dBERT	+Num	11.90	19.62
Adversary dBiDAF	+Num	29.80	36.619
Adversary dRoBERTa	+Num	13.10	21.16
Base SQuAD	+NE +Num	72.69	81.93
Adversary dBERT	+NE +Num	11.50	19.28
Adversary dBiDAF	+NE +Num	29.80	39.75
Adversary dRoBERTa	+NE +Num	12.70	20.13
Base SQuAD	+Num +NE	75.11	83.92
Adversary dBERT	+Num +NE	10.70	20.23
Adversary dBiDAF	+Num +NE	25.20	36.85
Adversary dRoBERTa	+Num +NE	12.00	21.29

Table 4. Named Entity + Numeric Subsets Fine-tuning the base model with dataset cartography Named Entity (NE) and Numeric (Num) subsets does not yield consistent performance level increases over base model.

Data	Base	EM	F1
Base SQuAD	+Amb+Hard	77.37	85.23
Adversary dBERT	+Amb+Hard	10.20	18.85
Adversary dBiDAF	+Amb+Hard	28.70	39.32
Adversary dRoBERTa	+Amb+Hard	13.10	21.49
Base SQuAD	+Hard+Amb	77.37	85.23
Adversary dBERT	+Hard+Amb	10.20	18.85
Adversary dBiDAF	+Hard+Amb	28.70	39.32
Adversary dRoBERTa	+Hard+Amb	13.10	21.49
Base SQuAD	+NE +Num	75.81	84.09
Adversary dBERT	+NE +Num	10.20	19.38
Adversary dBiDAF	+NE +Num	27.00	37.88
Adversary dRoBERTa	+NE +Num	12.50	21.70
Base SQuAD	+Num +NE	75.81	84.09
Adversary dBERT	+Num +NE	10.20	19.38
Adversary dBiDAF	+Num +NE	27.00	37.88
Adversary dRoBERTa	+Num +NE	12.50	21.70

Table 5.
Named Entity + Numeric Subsets (LR = 2e-6)
Ambiguous + Hard (LR = 7e-6)
Effective hyperparameter tuning can mitigate order-dependent effects in the fine-tuning process.